

Manual

HYDRA Interfacing Module to SAP PP via PP-PDC SAP-PPPDC 8.2

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1 HYDRA Interfacing Module to SAP PP via PP-PDC

Purpose

The PP-PDC (Production Planning – Plant Data Collection) is an interface of the SAP ECC production control. It replaces the former interface named communication channel 2 (KK2) and provides an extended functionality.

The interface has been implemented to connect to the MES. The MES applications record postings for operations that can then be uploaded. To check the plausibility of the data collection in the sub-system, the PP-PDC downloads operations (initial and delta download and delete function) and workplaces, deviation reasons and quantity units.

The implementation is performed using the [BAPI](#) technology. The SAP transceiver previously used has not been used. This procedure is the future standard for communication within an integrated SAP system architecture. Using the [BAPI](#) technology, the HYDRA connection is now performed on the level of business data.

Implementation notes

You use the PP-PDV interface for the following purposes:

- You have implemented the module Production Planning and Control in SAP ECC or R/3 and you use it for production orders
- You want to perform the shop floor control in HYDRA
- You want to perform the shop floor control in SAP, but you want to perform the progress postings on the production level
- Independent of the detailed scheduling that is performed, you want to perform an automatic connection of machines/systems and an automatic upload of the recorded quantities to SAP.
- You require actual times that are more precise to calculate the precise costs in SAP.

Integration

If you use the component SAP-PPDC, the orders/operations transferred with this component are used for a great number of further postings in HYDRA.

If the information provided via PP-PDC interface is not sufficient, you can use the component SAP-ISS HYDRA information interface for SAP PP to increase the information and to transfer additional data

Features

- Download of the released and planned order/operation data from SAP PP (initial and delta download and delete function)
- Recording of start/interruption/end of processing, start/interruption/end of setup, start/end of disturbance and validation checks
- Upload to SAP PP of the actual times and quantities recorded for a specific SAP time ticket
- Configurable assignment of the times recorded in HYDRA resource performance accounts to the SAP activity types.

2 PP-PDC integration in HYDRA

Purpose

If the ERP system and HYDRA are connected via the PP-PDC interface, the subsystem must be able to receive the [IDocs](#) generated by SAP ECC and to integrate them in the HYDRA process. HYDRA also generates IDocs from the recorded downloads/confirmations and transfers them to SAP ECC. SAP ECC triggers both workflows.

SAP provides several standard BAPIs / IDocs as part of the PP-PDC interface to establish communication with BDE subsystems. The following BAPIs / IDocs are used:

Download operations:

IDoc type:	PPCC2RECORDER01
Segment type:	E1BP_PP_PDC_OPERA2
BAPI segment:	BAPI_PP_PDC_OPERA2

Request uploads:

IDoc type:	PPCC2REQCONF
Segment type:	E1PPCC2REQCONF
BAPI segment:	BAPI_PP_PDC_PARAM

Upload time tickets:

IDoc type:	PPCC2PRETTICKET01
Segment type:	E1BP_PP_TIMETICKET
BAPI segment:	BAPI_PP_TIMETICKET

Unlike the previous interface, the communication channel 2, the new PP-PDC interface is completely based on BAPI and RFC. This means that the SAP transceiver is no longer required and that new RFC servers and clients are implemented in HYDRA. In HYDRA these new components communicate with SAP ECC and constitute the interface itself.

SAP ECC → HYDRA (download operation data)

HYDRA RFC server

The HYDRA RFC server logs in to the SAP gateway. The RFC server receives the incoming IDocs and stores them to the HYDRA database. Then the HYDRA process responsible for transferring data to the HYDRA data model is started.

BAPI / IDoc types

SAP ECC uses SAP workflow processes to trigger the download of data to HYDRA. Basically, the standard PP-PDC interface provides three IDoc structures. The HYDRA RFC server receives these IDoc structures: OPERA2 (initial / delta download), OPERA1 (deletion requests) and REQCONF (confirmation/upload requests).

Operation data (OPERA2)

An IDoc of the type PPCC2RECORDER01 transfers the operations. Depending on the entry in the segment E2PPCC2RECORDER000, it is either an initial download or a delta download. The segment E2BP_PP_PDC_OPERA2 includes operation data.

Deletion download (OPERA1)

An IDoc of the type PPCC2RECORDER01 transfers the operations. When it comes to a delta download, the segment E2BP_PP_PDC_OPERA1 can transfer the keys of the operations that are to be deleted.

Upload request (REQCONF)

An IDoc of the type PPCC2REQCONF transfers the request to upload time tickets. If this IDoc type is received, it is an upload request.

	OPERA2		OPERA1	REQCONF
	Initial download	Delta download	Deletion download	Upload request
IDoc type	PPCC2RECORDER01	PPCC2RECORDER01	PPCC2RECORDER01	PPCC2REQCONF01
Segment name	E2PPCC2RECORDER	E2PPCC2RECORDE R	E2PPCC2RECORDE R	E2PPCC2REQCONF
Field	INIT	INIT	INIT	REQTT
Entry	X			X

HYDRA → SAP ECC (upload)

SAP ECC asks for the upload of time tickets (L20 / L40). As an option, HYDRA can also start the upload at regular intervals, irrespective of an upload request.

HYDRA RFC client

The HYDRA RFC client uploads the time tickets. The client is started as part of a HYDRA workflow. At this time, the data is available ready for dispatch in the database.

Data is transferred asynchronously as IDoc. Once transferred, a defined SAP ECC workflow processes and posts the data.

Time tickets

An IDoc of the type PPCC2PRETTICKET01 of the standard PP-PDC interface transfers the time tickets. The segment E2BP_PP_TIMETICKET includes the time ticket data.

In general, the following SAP time ticket record types are uploaded:

- L20 – Partial confirmation (reporting part quantities/times)
- L40 – Final confirmation (reporting total quantities/times)

3 Behavior Depending on the Transfer Types

Initial download behavior

An initial download of the order and operation data will only take place when the complete system is commissioned and a first database is to be created in HYDRA. In addition, also other scenarios are possible by which the entire operations base will be replaced.

During an initial download all operations from SAP will be transferred that have at least the status "released" or that are not technically completed yet. This means that also (end) confirmed operations will be transferred but not imported in HYDRA.

When the initial download is received, the current operations data base in HYDRA will be deleted and be replaced by the new operations data. The current data will immediately be available.

An initial download will not only delete the order base existing in HYDRA but also all current times and quantities entered for orders and/or transactions.



In addition, all operations that are being executed will not be deleted. This means that they must be interrupted and/or terminated manually in HYDRA.

Conclusion: Any initial download during the operation of HYDRA must be used with greatest care and it would be useful to contact the MPDV Support beforehand.

For security reasons, the initial download function needs to be enabled explicitly as of the below-mentioned program version. Activation is performed by an INI configuration.

.lib\b_anr.dll

V8.1.1.326

Delta download behavior

A delta download creates new operations in HYDRA that will then be added to the database. Another function of the delta download is the modification of already transferred operations. Modifications of operations with the status "Running", "Finished" or "Deleted" are not allowed by the HYDRA default settings.



CUSTOMIZING information: Modifications of the operations depend on whether the flag "alterable order data" is set for the corresponding status in the status configuration.

If an operation cannot be modified to ensure the consistency in HYDRA (in general for running operations), this will be logged in HYDRA and be saved for error tracking purposes.

The basis for the delta download in SAP is the database table ORDCOM.

Deletion download behavior

A deletion download deletes those operations that are no longer necessary in the production process from the HYDRA database.

As with the modifications, the deletion result depends on the operation's status. Operations identified as "Running", "Interrupted", "Finished" or "Deleted", will not be deleted.

Moreover, the confirmation number (CONF_NO) will be checked to identify the orders via order, sequence and transaction. This prevents any accidental deletion of orders that are seemingly the same but which have different confirmation numbers.

Behavior when re-importing master data in SAP

When master data for an existing and released order that has already been transferred to HYDRA are re-imported in SAP this impacts also the interface to HYDRA.

When master data are re-imported, SAP will assign new confirmation numbers for the individual operations even though the order and operation numbers won't change. These are then transferred together with the next delta download to HYDRA. The confirmation/upload number will then be updated in HYDRA.



The update will also be made when the current status is "Running". In this case, however, ONLY the confirmation/upload no. will be changed and all other data (order quantity, scheduling, etc.) will not be updated.

Behavior during the technical completion in SAP

If an order is technically completed in SAP this will lead to a deletion download at the interface. This means that the order data in HYDRA will be deleted.

4 Download of operation data

Structure E2BP_PP_PDC_OPERA2

Field name	T	L	D	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	0	Logical system	Not used
CONF_NO	NUMC	10	0	Confirmation/upload number of the operation	Confirmation/upload number
ORDERID	CHAR	12	0	Order	According to configuration (*1)
SEQUENCE	CHAR	6	0	Sequence	According to configuration (*1)
OPERATION	CHAR	4	0	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	0	Suboperation	According to configuration (*1)
SUBSYSTEM_GROUPING	CHAR	3	0	Grouping of subsystem connection	Not used
MATERIAL	CHAR	18	0	Material	Order header, final item number, OP, item number
MATL_DESC	CHAR	40	0	Material description	Order header, final item description, OP item description
ROUT_QUAN_UNIT	UNIT	3	0	Quantity unit of the plan	Not used
ROUT_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
ORD_QUAN_UNIT	UNIT	3	0	Quantity unit for in-house production	Not used
ORD_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
BASE_QUAN_UNIT	UNIT	3	0	Base quantity unit	Base quantity unit; Entered in the order header for the first operation of an order. The operations "inherit" this base quantity unit from the order header. Please note: This unit must be the same for all operations of an order!
BASE_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used

Field name	T	L	D	Description	Usage in HYDRA
HDR_NOMINATOR	DEC	5	0	Numerator for the conversion to base quantity unit Please note: conversion factors from ORD_QUAN_UNIT into BASE_QUAN_UNIT. If OPER_QUAN_UNIT <> ORD_QUAN_UNIT wrong calculations might be the result!!!	Numerator for the conversion from primary to HYDRA base quantity unit
HDR_DENOMINATOR	DEC	5	0	Denominator for the conversion to the base quantity unit	Denominator for the conversion from primary to HYDRA base quantity unit
LEADING_ORDERID	CHAR	12	0	Order network: Leading order	Not used
SUPERIOR_ORDERID	CHAR	12	0	Order network: Directly superior order	Not used
SUPERIOR_SEQUENCE	CHAR	6	0	Order no.: Sequence of the directly superior order	Not used
SUPERIOR_OPERATION	CHAR	4	0	Order network: Operation of the directly superior order	Not used
REFERENCE_SEQUENCE	CHAR	6	0	Parallel sequence: Reference sequence of a sequence	Reference sequence
BRANCH_OPERATION	CHAR	4	0	Parallel sequence: Branch operation	Branch operation of the parallel sequence
RETURN_OPERATION	CHAR	4	0	Parallel sequence: Return operation	Return operation of the parallel sequence
OPER_DESCRIPTION	CHAR	40	0	Short text of operation	OP, operation designation
OPER_QUAN_UNIT	UNIT	3	0	Operation quantity unit	OP, quantity unit
OPER_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
OPER_QUANTITY	QUAN	13	3	Operation quantity	OP, target quantity (primary quantity)
PLANNED_SCRAP	QUAN	13	3	Scrap quantity of operation	OP, target scrap (primary quantity)
NOMINATOR	DEC	5	0	Numerator for the conversion from operation in planned quantity unit	Not used
DENOMINATOR	DEC	5	0	Denominator for the conversion from operation in planned quantity unit	Not used
UNDERDELIVERY_QUANTITY	QUAN	13	3	Quantity of the underdelivery tolerance in operation quantity unit	Conversion in percent: (UNDERDELIVERY_QUANTITY * 100) / OPER_QUANTITY

Field name	T	L	D	Description	Usage in HYDRA
CHECK_UNDERDELIVERY	CHAR	1	0	Check of the underdelivery tolerance in the order	" " No check "X" Error "W" Warning Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").
OVERDELIVERY_QUANTITY	QUAN	13	3	Quantity of the overdelivery tolerance in the operation quantity unit	Conversion in percent: $\frac{(\text{OVERDELIVERY_QUANTITY} * 100)}{\text{OPER_QUANTITY}}$
CHECK_OVERDELIVERY	CHAR	1	0	Check of the overdelivery tolerance in the order	" " No check "X" Error "W" Warning Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X").
MESSAGE_TYPE	CHAR	1	0	Message type for checking the operation sequence	Not used
USERFIELD_CH20_1	CHAR	20	0	User field for 20 characters (SAP USR00)	Usage in HYDRA see below *2)
USERFIELD_CH20_2	CHAR	20	0	User field for 20 characters (SAP USR01)	Usage in HYDRA see below *2)
USERFIELD_UNIT	UNIT	3	0	User field: Unit of quantity field (SAP USE04)	Not used
USERFIELD_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
USERFIELD_QUAN	QUAN	13	3	User field for the quantity (length 10,3) (SAP USR04)	Usage in HYDRA see below
ACTIVITY_UNIT_1	UNIT	3	0	Activity 1: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_1_ISO	CHAR	3	0	Activity 1: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_1	QUAN	13	3	Activity 1: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_2	UNIT	3	0	Activity 2: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_2_ISO	CHAR	3	0	Activity 2: ISO code of the quantity unit	Not used

Field name	T	L	D	Description	Usage in HYDRA
ACTIVITY_QUANTITY_2	QUAN	13	3	Activity 2: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_3	UNIT	3	0	Activity 3: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_3_ISO	CHAR	3	0	Activity 3: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_3	QUAN	13	3	Activity 3: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_4	UNIT	3	0	Activity 4: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_4_ISO	CHAR	3	0	Activity 4: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_4	QUAN	13	3	Activity 4: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_5	UNIT	3	0	Activity 5: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_5_ISO	CHAR	3	0	Activity 5: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_5	QUAN	13	3	Activity 5: Total activity quantity to be uploaded	Not used
ACTIVITY_UNIT_6	UNIT	3	0	Activity 6: Quantity unit of the activity to be uploaded.	Not used
ACTIVITY_UNIT_6_ISO	CHAR	3	0	Activity 6: ISO code of the quantity unit	Not used
ACTIVITY_QUANTITY_6	QUAN	13	3	Activity 6: Total activity quantity to be uploaded	Not used
CO_BUSPROC_1	CHAR	12	0	Business process: Identification	Not used
BUSPROC_UNIT_1	UNIT	3	0	Business proc.: quantity unit of the quantity to be uploaded.	Not used
BUSPROC_UNIT_1_ISO	CHAR	3	0	Business process: ISO code of the quantity unit	Not used
BUSPROC_QUANTITY_1	QUAN	13	3	Business process: quantity to be uploaded	Not used
WORK_CNTR	CHAR	8	0	Workplace	OP, machine (see note)
PLANT	CHAR	4	0	Plant	Used in confirmations/uploads

Field name	T	L	D	Description	Usage in HYDRA
EARL_SCHED_START_DATE	DATS	8	0	Earliest scheduled start time: Execution (date)	OP Earliest start date OP Planned start (date)
EARL_SCHED_START_TIME	TIMS	6	0	Earliest scheduled start time: Execution (time)	OP Earliest start time OP Planned start (time)
LATE_SCHED_FIN_DATE	DATS	8	0	Latest scheduled end date: Execution (date)	OP Latest end date OP Planned end (date)
LATE_SCHED_FIN_TIME	TIMS	6	0	Latest scheduled end time: Execution (time)	OP Latest end time OP Planned end (time)
SETUP_TIME_UNIT	UNIT	3	0	Unit of setup time	HYDRA converts into internal format
SETUP_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
SETUP_TIME	QUAN	7	1	Setup duration	OP Setup time (target BMK_7) *2)
PROCESS_TIME_UNIT	UNIT	3	0	Unit of processing time	HYDRA converts into internal format. The following units are supported: STD / H / HUR / HR
PROCESS_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
PROCESS_TIME	QUAN	7	1	Duration of processing	OP Processing time (target BMK_11) *2)
TEARDOWN_TIME_UNIT	UNIT	3	0	Unit of tear down time/retooling time	Not used
TEARDOWN_TIME_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used
TEARDOWN_TIME	QUAN	7	1	Teardown duration/retooling duration	OP, target teardown time/retooling time
STATUS	CHAR	5	0	Individual status of an object	Not used
INDICATOR_TT_OR_TE	CHAR	1	0	Time ticket or time event	Not used
INDICATOR_FIN_OR_PART	CHAR	1	0	Partial finish or end status	OPs with end status ("E") will not be transferred
CONFIRMED_YIELD	QUAN	13	3	Total yield uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_SCRAP	QUAN	13	3	Total scrap uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_REWORK	QUAN	13	3	Total rework uploaded/confirmed in operation quantity unit	Not used
CONFIRMED_ACTIVITY_1	QUAN	13	3	Activity 1: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_1	CHAR	1	0	Activity 1: Indicator: No remaining activity expected	Not used

Field name	T	L	D	Description	Usage in HYDRA
CONFIRMED_ACTIVITY_2	QUAN	13	3	Activity 2: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_2	CHAR	1	0	Activity 2: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_3	QUAN	13	3	Activity 3: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_3	CHAR	1	0	Activity 3: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_4	QUAN	13	3	Activity 4: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_4	CHAR	1	0	Activity 4: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_5	QUAN	13	3	Activity 5: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_5	CHAR	1	0	Activity 5: Indicator: No remaining activity expected	Not used
CONFIRMED_ACTIVITY_6	QUAN	13	3	Activity 6: Activity quantity already uploaded	Not used
NO_REMN_ACTIVITY_6	CHAR	1	0	Activity 6: Indicator: No remaining activity expected	Not used
CONFIRMED_BUS_PROC_1	QUAN	13	3	Quantity uploaded for the business process	Not used
NO_REMN_BUS_PROC_1	CHAR	1	0	No remaining quantity for the business process expected	Not used
CONFIRMED_WORK_CNTR	CHAR	8	0	Actual workplace	Not used
CONFIRMED_PLANT	CHAR	4	0	Actual plant	Not used

AK.xxx Field xxx in HYDRA order header

AG.xxx Field xxx in HYDRA operation

Structure E2BP_PP_PDC_OPERA1

The structure described in the following, controls the deletion process for already transferred production orders and/or their operations in the subsystem.

Field name	T	L	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	Logical system	Not used

Field name	T	L	Description	Usage in HYDRA
CONF_NO	NUMC	10	Confirmation/upload number of the operation	Confirmation/upload number
ORDERID	CHAR	12	Order	According to configuration (*1)
SEQUENCE	CHAR	6	Sequence	According to configuration (*1)
OPERATION	CHAR	4	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	Suboperation	According to configuration (*1)
SUBSYSTEM_GROUPING	CHAR	3	Grouping of subsystem connection	Can be restricted to a subsystem group.

Notes on the operation structure

HYDRA order number (*1)

The HYDRA order number consists of a configurable part of the SAP key fields ORDERID, SEQUENCE, OPERATION, SUB_OPER. The total length must not exceed 25 digits. If DOS terminals are used, the total length must not exceed 16 digits.

When the CHAR fields are transferred, the system converts blank characters to "0", since the barcode is also assigned "0" instead of blank characters.

Overview - usage of user fields

By default, the user-specific fields USERFIELD_CH20_1, USERFIELD_CH20_2 and USERFIELD_QUAN are processed in HYDRA as follows:

User field 01 (USERFIELD_CH20_1):

Places 1...10:

--> BDE cycle time in seconds per 1000 machine strokes in the format NUMC 10

Places 11...20:

--> not used in HYDRA

User field 02 (USERFIELD_CH20_2):

Places 1...10:

--> Premium default target te in seconds per 1000 pieces in the format NUMC 10

Places 11...20:

--> Premium default target tr in seconds in the format NUMC 10

User field 04 (USERFIELD_QUAN):

→ HYDRA partitioning (parts per cycle)

HYDRA partitioning

If the user field USERFIELD_QUAN has a value this value will be entered as the partitioning in the operation.

If this is not the case, the system implicitly assumes a partitioning of 1 and enters this value in the operation.

BDE cycle time

The system attempts to read the cycle time from user field 01 - places 1...10. If this is possible, the system enters this value as the target cycle time in the HYDRA operation.

If the system cannot identify the target cycle time as described above, then the system calculates it from the specified processing time (PROCESS_TIME) and the partitioning (TEILIGKEIT) values that might be specified.

Calculation basis (formulas):

Processing time OP = Target quantity OP * (target cycle time machine/ partitioning)

- Target cycle time machine (= BDE cycle time/ machine stroke)

- Target cycle time/ unit = target cycle time machine/ partitioning

=> Target cycle time machine= (processing time OP * Partitioning)/ target quantity OP

Standard times

In HYDRA the content of the field "processing time" (PROCESS_TIME) specifies the standard time for machine assignment (target for RPA 11).

In HYDRA the content of the field "Setup time" (SETUP_TIME) specifies the standard time for the machine setup (target for RPA 7).

Note on the workplace field WORK_CNTR:

As of version 6.5 HYDRA supports alphanumeric machine/workplace numbers. In this case, you can only use the Windows terminals CT 76x and CT 8xx for data collection.

Notes on all other alphanumeric fields:

HYDRA does not support specific special characters for all alphanumeric fields. These characters are: "%", "\", "/", "|" as you cannot enter these characters using shop floor terminals; the terminals and the MOC do not support these characters. Do not use the characters ";", " ", and " ' " since they are often interpreted as comment characters or separators and will thus lead to unwanted effects.

5 Key Fields / Supported Characters

General notes

In all alphanumeric fields, HYDRA does not support specific special characters. The following characters are not supported: "%", "ß", "*", "\", "/", "|", "_", "?". Reason: You cannot enter these characters on the shop floor terminals or the clients do not support these characters.

You must not use the characters ";" (semicolon), "," (comma) and "'" (apostrophe) because they are often interpreted as comment characters or separators and can lead to unwanted results.

Workplace/machine numbers (resources of type "MNR")

Workplace/machine numbers and numbers of capacity/machine groups are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 8, left-aligned.

When you create or copy a workplace, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30_{hex} - 39_{hex}](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41_{hex} - 5A_{hex}](#))
- "-" ([US-ASCII 2D_{hex}](#))

Lower case letters are automatically converted to upper case letters when a new workplace is created.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

The entry "SYSTEM" as workplace/machine number is reserved for HYDRA and may not be used.

It is possible to overwrite the valid characters for the workplace/machine numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

Field	Value
Name	INPUT
Section	PATTERN
Key	MNR
Value	^(?!SYSTEM) ([A-Z0-9(){}~^#+!\$. _%-]+)\$
Active	☒

Minimum requirement: b_mnr.dll version 8.1.1.102

Resource numbers (resources of type <> "MNR")

Resource numbers are interpreted as alphanumeric values. Alphanumeric field with a maximum length of 20; left-aligned.

When you create or copy a resource, the system checks if the characters used are allowed. The following characters are allowed:

- Numbers "0" to "9" ([US-ASCII 30_{hex} - 39_{hex}](#))
- Letters "A" - "Z" (upper case letters - [US-ASCII 41_{hex} - 5A_{hex}](#))
- Umlauts "Ä", "Ö", "Ü" ([Extended ASCII C4_{hex}, D6_{hex}, DC_{hex}](#))
- "-" ([US-ASCII 2D_{hex}](#))

You may not use umlauts or special characters (e.g. "%", "ß", "*", "\", "/", "|", "_", "?") because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

You must not use blanks. If required, you must prefix the numbers by leading zeros ("0").

Lower case letters are automatically converted to upper case letters when a new resource is created.

It is possible to overwrite the valid characters for the resource numbers in the INI configuration. To this end, you must specify the valid characters as a regular expression (in brackets).

Field	Value
Name	INPUT
Section	PATTERN
Key	RES
Value	^([0-9A-ZÄÖÜ(){}~^#+!\$._%-]+)\$
Active	☒

Minimum requirement: b_res.dll version 8.1.1.117

HYDRA order number

There are some differences with respect to the order number in the HYDRA data model and the interface.

Order number

The order number (field AUNR) contains the actual order number as it is known in the ERP system and transferred to HYDRA. The order number is specified in the HYDRA basic settings; by default, this number has a length of 8 characters.

Operation number

The operation number (field AGNR) clearly identifies a defined process step of an order. The operation number is specified in the HYDRA basic settings; by default, this number has a length of 4 characters.

MES order number

The MES order number (field ANR) combines the order and the operation number and sometimes also the sequence number from the ERP system (if licensed). Its length therefore results from the total of the separate number lengths.

The total length must not exceed 25 digits. If DOS terminals are used, the total length must not exceed 16 digits.

Note the following for the order or operation number:

- Preferably only use the numbers "0" to "9" ([US-ASCII 30_{hex}](#) - [39_{hex}](#)).
- If you use letters, only the characters "A" - "Z" (upper case - [US-ASCII 41_{hex}](#) - [5A_{hex}](#)) and "-" ([US-ASCII 2D_{hex}](#)) are allowed. Do not use lower case letters.
- You may not use blanks in the numbers. The order or operation numbers must have the specified number of digits with the characters "0" to "9" or "A" to "Z". If required, you must prefix the numbers by leading zeros ("0").
- HYDRA does not support any umlauts, blank or special characters (see section *General notes*) for the order or operation number because you cannot enter these characters on the shop floor terminals or because the clients do not support these characters.

6 Download of Master Data

Download of the work centers

SAP allows to download workcenters to run plausibility checks in the subsystem. These work centers can be transferred to HYDRA.

When these work centers will be transferred, the system will check for the respective SAP work center in the HYDRA configuration order standard whether an entry is stored for this SAP work center by which this entry will be transferred as HYDRA workplace. In the beginning, the workplace will be created with the responsibility area "SAP" by the user "SAP". Due to this it is necessary to create the user "SAP" and to assign it then to the responsibility area "SAP". During the adoption workplaces will be inserted or updated. Deletions are not possible.

Since work centers are only unique within a plant in SAP; it must be configured in HYDRA for which plant you wish to adopt the work centers. This configuration is made in the HYDRA Ini-configuration.

Name: PP-PDC

Section: PPCC2RECWORKCENTER

Key: Plant, for which the work centers are to be adopted, e.g. 1000

Value: <BLANK> (no value must be entered here)

Field name	T	L	D	Description	Usage in HYDRA
SOURCE_SYS	CHAR	10	0	Logical system	Not used
WORK_CNTR	CHAR	8	0	Work center	HYDRA machine/ workplace
PLANT	CHAR	4	0	Plant	Plant
SUBSYSTEM_GROUPING	CHAR	3	0	BDE group	Transfer to HYDRA
WORK_CNTR_DESCR	CHAR	40	0	Short text on the work center	Comment
VALID_START	DATS	8	0	Start of validity of the current cost center link	The data record will only be adopted when the current date is within this interval.
VALID_END	DATS	8	0	End of validity of the current cost center link	The data record will only be adopted when the current date is within this interval.
CO_AREA	CHAR	4	0	Controlling area of the current cost center link	Not used
COST_CNTR	CHAR	10	0	Cost center of the current cost center link	Cost center
ACTI1_TEXT	CHAR	20	0	Activity 1: Activity text	Not used
ACTI1_UNIT	UNIT	3	0	Activity 1: Activity unit	Not used
ACTI1_UNIT_ISO	CHAR	3	0	Activity 1: ISO code of the quantity unit	Not used
NOACTI1	CHAR	1	0	Activity 1: Indicator: Do not show activity text	Not used
RECORD_GRP1	NUMC	1	0	Activity 1: Record type group	Not used
ACTI2_TEXT	CHAR	20	0	Activity 2: Activity text	Not used

Field name	T	L	D	Description	Usage in HYDRA
ACTI2_UNIT	UNIT	3	0	Activity 2: Activity unit	Not used
ACTI2_UNIT_ISO	CHAR	3	0	Activity 2: ISO-code of the quantity unit	Not used
NOACTI2	CHAR	1	0	Activity 2: Indicator: Do not show activity text	Not used
RECORD_GRP2	NUMC	1	0	Activity 2: Record type group	Not used
ACTI3_TEXT	CHAR	20	0	Activity 3: Alternative activity text	Not used
ACTI3_UNIT	UNIT	3	0	Activity 3: Activity unit	Not used
ACTI3_UNIT_ISO	CHAR	3	0	Activity 3: ISO code of the quantity unit	Not used
NOACTI3	CHAR	1	0	Activity 3: Indicator: Do not show activity text	Not used
RECORD_GRP3	NUMC	1	0	Activity 3: Record type group	Not used
ACTI4_TEXT	CHAR	20	0	Activity 4: Activity text	Not used
ACTI4_UNIT	UNIT	3	0	Activity 4: Activity unit	Not used
ACTI4_UNIT_ISO	CHAR	3	0	Activity 4: ISO code of the quantity unit	Not used
NOACTI4	CHAR	1	0	Activity 4: Indicator: Do not show activity text	Not used
RECORD_GRP4	NUMC	1	0	Activity 4: Record type group	Not used
ACTI5_TEXT	CHAR	20	0	Activity 5: Alternative activity text	Not used
ACTI5_UNIT	UNIT	3	0	Activity 5: Activity unit	Not used
ACTI5_UNIT_ISO	CHAR	3	0	Activity 5: ISO code of the quantity unit	Not used
NOACTI5	CHAR	1	0	Activity 5: Indicator: Do not show activity text	Not used
RECORD_GRP5	NUMC	1	0	Activity 5: Record type group	Not used
ACTI6_TEXT	CHAR	20	0	Activity 6: Activity text	Not used
ACTI6_UNIT	UNIT	3	0	Activity 6: Activity unit	Not used
ACTI6_UNIT_ISO	CHAR	3	0	Activity 6: ISO code of the quantity unit	Not used
NOACTI6	CHAR	1	0	Activity 6: Indicator: Do not show activity text	Not used
RECORD_GRP6	NUMC	1	0	Activity 6: Record type group	Not used
CO_BUSPROC	CHAR	12	0	Business process	Not used
CO_BUSPROC_NAME	CHAR	20	0	General designation of the business process	Not used
COST_DRIVER	UNIT	3	0	Activity unit CO-ABC on the business process	Not used
COST_DRIVER_ISO	CHAR	3	0	Activity unit CO-ABC according to ISO on the business process	Not used

Download of the deviation reasons

SAP allows to download deviation reasons to run plausibility checks in the subsystem. These deviation reasons can be transferred to HYDRA.

When the deviation reasons are transferred, the system will first create a reason text with the number transferred from SAP. Then a reason together with the number transferred from SAP and referring to the created reason text will be created.

In the beginning, the reason texts and reasons are created by the "SAP" user. This user must exist in the system as HYDRA user. During the transfer reasons/ reason texts will be inserted or updated. Deletions are not possible.

Since deviation reasons are only unique within a plant in SAP, it must be configured in HYDRA for which plant you wish to transfer the deviation reasons. This configuration is made in the HYDRA Ini-configuration.

Name: PP-PDC

Section: DIFFE2

Key: Plant, for which the reasons are to be transferred, e.g. 1000

Value: Reason type

A	Scrap
N	Rework
P	Problem quantity
G	Yield
L	Batch logs

Field name	Type	L		Meaning	Usage in HYDRA
WERKS	CHAR	4		Plant	Plant
REASON	CHAR	4		Variation cause	ID Reason Text ID Reason
GRDTX	CHAR	25		Text stating the reasons of the deviation	Scrap reason
SOURCE_SYS	CHAR	10		Logical system	Not used

Download of generally applicable units

SAP allows to download generally applicable units to run plausibility checks in the subsystem. These generally applicable units can be transferred to HYDRA.

In the beginning, the units are created by the "SAP" user. This user must exist in the system as HYDRA user. During the transfer the units will be inserted or updated. Deletions are not possible.

During the transfer the SI-units will not be identified and the ISO code for the units cannot be transferred.

Field name	Type	L	D	Meaning	Usage in HYDRA
MSEHI	CHAR	3		Quantity unit (internal key)	Not used
MSEHE	CHAR	3		Quantity unit (external key)	Unit
NENNR	CHAR	10		Denominator for the conversion into SI-unit	Not used
ZAEHL	CHAR	10		Numerator for the conversion into SI-unit	Not used
MSSIE	CHAR	3		SI-unit (internal key)	Not used
MSEHL	CHAR	25		Text on the quantity unit	Text on the quantity unit

Field name	Type	L	D	Meaning	Usage in HYDRA
ANDEC	CHAR	3		Number of decimal places	Not used
SOURCE_SYS	CHAR	10		Logical system	Not used

Download of material-dependent units

SAP allows to download material-dependent units to run plausibility checks in the subsystem. These units can be transferred to HYDRA.

In the beginning, the units are created by the "SAP" user. This user must exist in the system as HYDRA user. During the transfer the units will be inserted or updated. Deletions are not possible.

Field name	Type	L	D	Meaning	Usage in HYDRA
MATNR	CHAR	18		Material	Material
MEINH	CHAR	3		Alternative quantity unit of the material (internal key)	Unit of
MEINS	CHAR	3		Base quantity unit of the material (internal key)	Unit by
UMREZ	CHAR	7		Numerator for the conversion of alternatives in stock keeping unit	Numerator
UMREN	CHAR	7		Denominator for the conversion of alternatives in stock keeping unit	Denominator
SOURCE_SYS	CHAR	10		Logical system	Not used

7 Upload of SAP time tickets

Overview

Record types and activities supported by HYDRA

HYDRA-BDE uploads the following record types in relation to time tickets to SAP R/3 PP.

Record type	Meaning in SAP	Triggering HYDRA action
L20	Partial completion of time ticket	Orders are interrupted automatically or manually via the shop floor client or office client.
L40	End of time ticket	Completion of an order via the shop floor client or office client.



If HYDRA MPL is active, an L20 partial completion is generated and transferred to SAP for every generated output batch (output batch changed) in addition to the SAP time tickets for interrupting or logging off the OP.

Confirmation/upload structure (E2BP_PP_TIMETICKET)

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
CONF_NO	NUMC	10	0	Confirmation/upload number of the operation	Confirmation/upload number as specified	1	10
ORDERID	CHAR	12	0	Order	SAP order according to specifications	11	22
SEQUENCE	CHAR	6	0	Sequence	SAP sequence according to specifications	23	28

1 The indicated number of characters is calculated based on the export length of the GI transaction 31 in SAP and can be used as reference for HYDRA developers.

2 See footnote of column "From"

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
OPERATION	CHAR	4	0	Operation	SAP operation according to specifications	29	32
SUB_OPER	CHAR	4	0	Suboperation	SAP suboperation according to specifications	33	36
CAPA_CATEGORY	CHAR	3	0	Capacity category	SAP capacity category according to specifications	37	39
SPLIT	INT1	3	0	Split number	000, as no specification possible	40	42
FIN_CONF	CHAR	1	0	Partial/final confirmation/upload (not interpreted)	Not used	43	43
CLEAR_RES	CHAR	1	0	Clearing open reservations	Use the parameter /CLEAR_RES to assign an "X" to this field for an L40 posting.	44	44
POSTG_DATE	DATS	8	0	Posting date	Shift date of the HYDRA posting record	45	52
DEV_REASON	CHAR	4	0	Reason for the deviation	This field includes the scrap reason if the options "report part quantities" or "upload of scrap including reason" are enabled for the order type .	53	56
CONF_TEXT	CHAR	40	0	Confirmation/upload text	Not used Exception: batch numbers are entered here if H records are uploaded/confirmed	57	96
PLANT	CHAR	4	0	Plant	Specified plant	97	100
WORK_CNTR	CHAR	8	0	Workplace	According to configuration - see below	101	108
RECORDTYPE	CHAR	3	0	Record type of the upload/confirmation	L20 or L40	109	111
CONF_QUAN_UNIT	UNIT	3	0	Quantity unit of confirmation/upload	OP target quantity unit (primary quantity unit)	112	114
CONF_QUAN_UNIT_ISO	CHAR	3	0	ISO code of quantity unit of confirmation/upload	Not used	115	117
YIELD	QUAN	13	3	Yield	Yield in primary quantity unit	118	132
SCRAP	QUAN	13	3	Scrap quantity	Scrap in primary quantity unit	133	147
REWORK	QUAN	13	3	Rework quantity	Rework quantity in primary quantity unit	148	162

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
CONF_ACTI_UNIT1	UNIT	3	0	Activity 1: quantity unit of activity quantity	According to configuration	163	165
CONF_ACTI_UNIT1_ISO	CHAR	3	0	Activity 1: ISO code of the quantity unit	Not used	166	168
CONF_ACTIVITY1	QUAN	13	3	Activity 1: activity quantity	According to configuration	169	183
NO_REMN_ACTI1	CHAR	1	0	Activity 1: No remaining activity expected	Not used	184	184
CONF_ACTI_UNIT2	UNIT	3	0	Activity 2: quantity unit of activity quantity	According to configuration	185	187
CONF_ACTI_UNIT2_ISO	CHAR	3	0	Activity 2: ISO code of the quantity unit	Not used	188	190
CONF_ACTIVITY2	QUAN	13	3	Activity 2: activity quantity	According to configuration	191	205
NO_REMN_ACTI2	CHAR	1	0	Activity 2: No remaining activity expected	Not used	206	206
CONF_ACTI_UNIT3	UNIT	3	0	Activity 3: quantity unit of activity quantity	According to configuration	207	209
CONF_ACTI_UNIT3_ISO	CHAR	3	0	Activity 3: ISO code of the quantity unit	Not used	210	212
CONF_ACTIVITY3	QUAN	13	3	Activity 3: activity quantity	According to configuration	213	227
NO_REMN_ACTI3	CHAR	1	0	Activity 3: No remaining activity expected	Not used	228	228
CONF_ACTI_UNIT4	UNIT	3	0	Activity 4: quantity unit of activity quantity	According to configuration	229	231
CONF_ACTI_UNIT4_ISO	CHAR	3	0	Activity 4: ISO code of the quantity unit	Not used	232	234
CONF_ACTIVITY4	QUAN	13	3	Activity 4: activity quantity	According to configuration	235	249
NO_REMN_ACTI4	CHAR	1	0	Activity 4: No remaining activity expected	Not used	250	250
CONF_ACTI_UNIT5	UNIT	3	0	Activity 5: quantity unit of activity quantity	According to configuration	251	253
CONF_ACTI_UNIT5_ISO	CHAR	3	0	Activity 5: ISO code of the quantity unit	Not used	254	256

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
CONF_ACTIVITY5	QUAN	13	3	Activity 5: activity quantity	According to configuration	257	271
NO_REMN_ACTI5	CHAR	1	0	Activity 5: No remaining activity expected	Not used	272	272
CONF_ACTI_UNIT6	UNIT	3	0	Activity 6: quantity unit of activity quantity	According to configuration	273	275
CONF_ACTI_UNIT6_ISO	CHAR	3	0	Activity 6: ISO code of the quantity unit	Not used	276	278
CONF_ACTIVITY6	QUAN	13	3	Activity 6: activity quantity	According to configuration	279	293
NO_REMN_ACTI6	CHAR	1	0	Activity 6: No remaining activity expected	Not used	294	294
CONF_BUS_PROC_UNIT1	UNIT	3	0	Business process: quantity unit of business process quantity	Not used	295	297
CONF_BUS_PROC_UNIT1_ISO	CHAR	3	0	Business process: ISO code of the quantity unit	Not used	298	300
CONF_BUS_PROC1	QUAN	13	3	Business process: business process quantity	Not used	301	315
NO_REMN_BUS_PROC1	CHAR	1	0	Business process: no remaining quantity expected	Not used	316	316
EXEC_START_DATE	DATS	8	0	Date when "starting execution"	Start time of the confirmed/uploaded posting record	317	324
EXEC_START_TIME	TIMS	6	0	Time when "starting execution"	Start time of the confirmed/uploaded posting record	325	330
SETUP_FIN_DATE	DATS	8	0	Date when "finishing setup"	Not used	331	338
SETUP_FIN_TIME	TIMS	6	0	Time when "finishing setup"	Not used	339	344
PROC_START_DATE	DATS	8	0	Date when "starting processing"	Not used	345	352
PROC_START_TIME	TIMS	6	0	Time when "starting processing"	Not used	353	358
PROC_FIN_DATE	DATS	8	0	Date when "finishing processing"	Not used	359	366
PROC_FIN_TIME	TIMS	6	0	Time when "finishing processing"	Not used	367	372

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
TEARDOWN_START_DATE	DATS	8	0	Date when "starting retooling/teardown"	Not used	373	380
TEARDOWN_START_TIME	TIMS	6	0	Time when "starting retooling/teardown"	Not used	381	386
EXEC_FIN_DATE	DATS	8	0	Date when "finishing execution"	End time of the confirmed/uploaded posting record	387	394
EXEC_FIN_TIME	TIMS	6	0	Time when "finishing execution"	End time of the confirmed/uploaded posting record	395	400
FCST_FIN_DATE	DATS	8	0	Date of the forecast "end of execution"	Not used	401	408
FCST_FIN_TIME	TIMS	6	0	Time of the forecast "end of execution"	Not used	409	414
STD_UNIT1	UNIT	3	0	Default value 1: quantity unit	Not used	415	417
STD_UNIT1_ISO	CHAR	3	0	Default value 1: ISO code of quantity unit	Not used	418	420
FORECAST_STD_VAL1	QUAN	9	3	Default value 1: forecast default value	Not used	421	431
STD_UNIT2	UNIT	3	0	Default value 2: quantity unit	Not used	432	434
STD_UNIT2_ISO	CHAR	3	0	Default value 2: ISO code of quantity unit	Not used	435	437
FORECAST_STD_VAL2	QUAN	9	3	Default value 2: forecast default value	Not used	438	448
STD_UNIT3	UNIT	3	0	Default value 3: quantity unit	Not used	449	451
STD_UNIT3_ISO	CHAR	3	0	Default value 3: ISO code of quantity unit	Not used	452	454
FORECAST_STD_VAL3	QUAN	9	3	Default value 3: forecast default value	Not used	455	465
STD_UNIT4	UNIT	3	0	Default value 4: quantity unit	Not used	466	468
STD_UNIT4_ISO	CHAR	3	0	Default value 4: ISO code of quantity unit	Not used	469	471
FORECAST_STD_VAL4	QUAN	9	3	Default value 4: forecast default value	Not used	472	482

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
STD_UNIT5	UNIT	3	0	Default value 5: quantity unit	Not used	483	485
STD_UNIT5_ISO	CHAR	3	0	Default value 5: ISO code of quantity unit	Not used	486	488
FORCAST_STD_VAL5	QUAN	9	3	Default value 5: forecast default value	Not used	489	499
STD_UNIT6	UNIT	3	0	Default value 6: quantity unit	Not used	500	502
STD_UNIT6_ISO	CHAR	3	0	Default value 6: ISO code of quantity unit	Not used	503	505
FORCAST_STD_VAL6	QUAN	9	3	Default value 6: forecast default value	Not used	506	516
FORCAST_BUS_PROC_UNIT1	UNIT	3	0	Business process: quantity unit of forecast Remaining quantity	Not used	517	519
FORC_BUS_PROC_UNIT1_ISO	CHAR	3	0	Business process: ISO code of the quantity unit	Not used	520	522
FORCAST_BUS_PROC_VAL1	QUAN	13	3	Business process: forecast remaining quantity	Not used	523	537
PERS_NO	NUMC	8	0	Personnel number	Not used	538	545
TIMEID_NO	NUMC	8	0	Time recording ID card number	According to configuration	546	553
WAGETYPE	CHAR	4	0	Wage type	Not used	554	557
SUITABILITY	CHAR	2	0	Suitability	Not used!	558	559
NO_OF_EMPLOYEE	DEC	5	2	Number of employees	Not used	560	566
WAGEGROUP	CHAR	3	0	Wage group	Not used	567	569
BREAK_UNIT	UNIT	3	0	Unit of break time	Not used	570	572
BREAK_UNIT_ISO	CHAR	3	0	ISO code of quantity unit	Not used	573	575
BREAK_TIME	QUAN	9	3	Uploaded/confirmed break time	Not used	576	586

Field name	T	L	D	Meaning	Usage in HYDRA	From 1	To2
EX_CREATED_BY	CHAR	12	0	External person creating the confirmation/upload	Not used	587	598
EX_CREATED_DATE	DATS	8	0	External date of entering the confirmation/upload	Not used	599	606
EX_CREATED_TIME	TIMS	6	0	External time of entering the confirmation/upload	Not used	607	612
TARGET_ACTI1	CHAR	1	0	Indicator: identify target activity 1	Is assigned to "X" when calculating activities in SAP	613	613
TARGET_ACTI2	CHAR	1	0	Indicator: identify target activity 2	Is assigned to "X" when calculating activities in SAP	614	614
TARGET_ACTI3	CHAR	1	0	Indicator: identify target activity 3	Is assigned to "X" when calculating activities in SAP	615	615
TARGET_ACTI4	CHAR	1	0	Indicator: identify target activity 4	Is assigned to "X" when calculating activities in SAP	616	616
TARGET_ACTI5	CHAR	1	0	Indicator: identify target activity 5	Is assigned to "X" when calculating activities in SAP	617	617
TARGET_ACTI6	CHAR	1	0	Indicator: identify target activity 6	Is assigned to "X" when calculating activities in SAP	618	618
TARGET_BUS_PROC1	CHAR	1	0	Indicator: identify target quantity of business process	Not used	619	619
EX_IDENT	CHAR	32	0	External key of the upload/confirmation (GUID)	Distinct key identifying the upload/confirmation Reference from HYDRA table ADE_PROTOKOLL See the notes at the end of the table.	620	651
LOGDATE	DATS	8	0	Logical date	Date of the upload/confirmation	652	659
LOGTIME	TIMS	6	0	Logical time	Time of the upload/confirmation	660	665

Remarks on selected fields

EX_IDENT

The field EX_IDENT is assigned the value from the *verweis* column of the *ade_protokoll* table. The field displays a continuous database serial and thus guarantees uniqueness within a DB instance.

The field is populated with the value of the *verweis* column and leading zeros (left-aligned) to reach full length. Example: the *verweis* 4711 leads to "00000000000000000000000000004711".

You can use the program parameter of the upload program myerprck.exe/out to assign a prefix. The prefix allows you to use multiple HYDRA systems with one SAP instance. To do so, you can use the program parameter "[IDENT_PRAEFIX](#)".



You cannot upload part quantities (L20/L40) and record data at the same time via the total quantity counter at MDE machines, as SAP cannot process negative quantities by default. This type of collection can result in negative quantity postings for yield when OPs are finished.

This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations).



The sign is located at the end of fields of the type "QUAN".
Example: 0000012345.432+

8 SAP Order Sequencing

Overview

HYDRA menu	System administration → MES Link Enabling → SAP order sequencing
FEDRA menu	System administration → MES Link Enabling → SAP order sequencing
Transaction code	mleoss
Function authorization	mleoss.*

Purpose

Use the *SAP order sequencing* to control how the workplaces specified by SAP are transferred to the system. You can choose from the following options:

- The workplace transferred from SAP is interpreted as HYDRA group and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is directly planned for the machine.

This decision either affects the workplace or the entire system.

Integration

Diverse interfaces use these configurations to transfer orders from SAP.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Key

Use this field to specify whether the entry applies to a specific order type or a machine.

In general, the configuration refers to a workplace.

Value

If you selected *order type* as the key, use this field to enter the order type the configuration applies for.

If you selected *machine* as the key:

- Enter a separate workplace in the *value* field if the configuration should apply for a specific workplace.
- Enter the value SYSTEM if you want the configuration to apply for the entire system.

Configurations referring to a specific machine take priority over the SYSTEM setting.



You can make a system entry for the majority of machines/workplaces/work centers and exceptions may be configured specifically.

Detailed planning

- **G**

"Transfer the SAP workplace as HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **M**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **N**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, and plan the operation for the workplace."

- **Rule 1 - Rule 9**

Use the rules 1 to 9 for customer-specific transfer logics, which will be implemented as part of the project.

9 SAP Upload

Overview

HYDRA menu	System administration → MES Link Enabling (MLE) → SAP upload
FEDRA menu	System administration → MES Link Enabling (MLE) → SAP upload
Transaction code	mlecos
Function authorization	mlecos.*

Purpose

You use this function to configure the transfer from the resource performance accounts to the SAP activity types. You can make this setting for a single workplace or globally for the entire system.

Integration

This setting is used when the SAP time tickets are uploaded to SAP PP via the PP-PDC interface.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Workplace

If a machine is specified, the configuration is applied for this machine (system workplace). If you enter "SYSTEM", this is a global setting and is used for all machines, except for machine with an own, specific entry.

Activity 1 – Activity 6

These fields specify which combination of time accounts kept in the system is uploaded. Possible values:

- RPA1 – RPA12
Resource performance accounts of machine from U/E record
- PRPA1 – PRPA12
Resource performance accounts of persons from U/E record (the upload of times from the resource performance accounts of persons [must be activated separately](#).)
- P_DAUER
Labor utilization

To post the total of several time accounts to an activity field, the accounts are listed separated by the "+" sign.

Example: RPA1+RPA2+RPA3+RPA4+RPA5+RPA6+RPA7+RPA8+RPA9+RPA10+RPA11

You can also upload quantity accounts, not only time accounts. Possible values are:

GUT (yield),

AUS (scrap).

To post the total of several quantity accounts to an activity field, the accounts are listed separated by the "+" sign.

Example: GUT+AUS

Unit 1 – Unit 6

Enter the relevant upload unit in these fields. Valid values:

Hours	H, HUR, HR, STD
Minutes	MIN
Seconds	SEC
One tenth of an hour (6 min)	ZE

Target activity 1 – Target activity 6

If you check this option, you activate the calculation of the activities using the number of pieces defined in SAP for the activity type. If the option is enabled, the activities are no longer transferred to SAP. If you disable the option, the transfer is enabled again. Also this setting can be made globally for the entire system or for a specific machine only. If the fields are checked, the fields TARGET_ACTI1 – TARGET_ACTI6 in the upload structure "TIMETICKET" are equally assigned an "X".

10 MYERPRCK - Program Parameters

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Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

Parameters	Meaning/use	Relevant interfaces	Productive release
Program parameters to control processing:			
/MESTYP=XXXX	The parameter MESTYP defines the structure to be generated.	All	Yes
/GRP=XXXX	The grouping type specifies the criterion by which uploads should be grouped. Possible values: PLANT --> Groups by plant	Requires customizations	Requires customizations
/V=sssss	Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time. Use the parameter /V=sssss (sssss =	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	delay time in seconds) to activate the above described delay when the upload program is called. Examples: <pre>myerprck.exe/out /V=3600</pre> The system only uploads postings that are older than one hour.	SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	
/BIS=DDMMYYHHMM /BIS=HHMM /TILLDATE=MM/DD/YYYY /TILLTIME=sec after midnight	Use the parameter /BIS=DDMMYYHHMM (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day. <pre>Myerprck.exe /BIS=2505110600</pre> This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011. <pre>Myerprck.exe /BIS=0600</pre> This parameter uploads postings that were recorded until 06:00 a.m. of the current day.	EIS-ERP EIS-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TZ=+/-sssss	Use the parameter /TZ=+/-sssss to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPPDC interface according to its specifications.	SAP-PPPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KST=XXX	Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center. Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/CLEAR_RES	Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.	SAP-PPDC	Yes
/NEG_MENGE	By default, quantities (L20/L40) cannot	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPDC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished. This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.		
/LA_MNR	The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM. The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type. Then the system uses the machine to identify the activity type from the activity types kept in HYDRA.	SAP-PMCC3	Yes
/IDENT_PRAEFIX=	In the upload structure of the SAP-PPDC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field. You can add a prefix to the EX_IDENT field to differentiate between uploads	SAP-PPDC SAP-PPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	from various HYDRA subsystems connected to one SAP instance. <u>Example:</u> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> The prefix may only include hexadecimal characters: A –H und 0 – 9.		
/ABZEICH=XX	While customizing the order type, you can specify that only signed data records are uploaded. Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.	EIS-ERP EIS-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TRANSFER=	Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/NOTTRANSFER=XXX	Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SEK	The EIS-ERP interface uploads the times of resource performance accounts in hours. In particular with very short lead times this may effect that logon times are cut off by a conversion into hours. Use this program parameter to upload times in seconds.	EIS-ERP ESI-XPPS	Yes
/RMTYP=	When customizing the order type, you can assign an upload type to the order type. Use this program parameter to only upload data records of this upload type. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KAT=	When customizing the order type, you can connect the order type with a category. Use the program parameter /KAT= to only upload data records of this category. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SART=	The system only uploads ADE log postings of the specified record type. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the corresponding uploads when customizing the order type. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /SART=A /SART=E</pre> ⇒ The system only uploads A and E records.	EIS-ERP ESI-XPPS SAP-PPPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/NOLOCK	When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,	All	Requires customizations

Parameters	Meaning/use	Relevant interfaces	Productive release
	the upload is not carried out. You can use this program parameter to prevent this check. Set this parameter, in particular, if the upload is not based on the table ade_protokoll.		
/EINH_CC34	The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter "/EINH_CC34" to upload the data in other units. The following units are supported: Hours: H, HUR, STD Minutes: MIN Seconds: SEC <u>Example:</u> Myerprck.exe /EINH_CC34=HUR The system uploads the recorded times in the unit "HUR" (hours).	SAP-PMCC3 SAP-PSCC4	Yes
/SDAT_STORNO	The SAP-PPPDCC interface transfers the change date along with the correction records. Use this program parameter to upload the initially collected shift date instead.	SAP-PPPDCC	Yes
/NORFC_STORNO	The SAP-PPPDCC interface transfers the cancellation records via sRFC. Use the program parameter to transfer the data in the IDoc format to SAP. To do	SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	so, inbound processing must be implemented in SAP. The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.		
/PI	If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly. Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).	SAP-PPDC SAP-PMCC3 SAP-PSCC4	Yes
/INDEX_TMP_TABLE	Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected. To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.	All	Requires customizations
/UE_PARAMS=	Program parameter for the stand-alone user exit processing (DD format).	Various	Yes
/NOSTORNO	Use this program parameter to prevent cancellation records from being uploaded. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the	All	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	corresponding uploads when customizing the order type.		
/RECALC_NEG_YIELD	Use this parameter to offset negative yield with already posted positive uploads.	SAP-PPPDCC	Requires customizations
Program parameters to use the SIGUSR communication:			
/LOGGING	Use this program parameter to activate communication from the database table HYD_LOGGING. To do so, a customization might be required.	INDIVIDUAL CASE	Yes
/WAIT_SIGUSR1=XX	The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.	INDIVIDUAL CASE	Yes
/PEEK_SIGUSR1=XX	Use this parameter to delay execution of an action triggered by the SIGUSR communication. The delay time is entered in seconds for this parameter. The program interprets this time as follows: If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds. If in a specific case, triggers would indeed arrive every second then the	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.		
/SEND_SIGUSR1=	This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication. Specify the process/program WITHOUT file extension.	INDIVIDUAL CASE	Yes
/COUNT_SIGUSR1=XX	Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance. Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1. Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	the DB. Benefits: If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.		
Program parameters for debugging/ tracing/ testing/ logging purposes:			
/ONLYERR	This program parameter specifies that system log entries are only created if an error occurred during uploading. This reduces the entries in the system log.	All	Yes
/SIM	The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").	All	No
/SIMULATION	The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).	All	No

11 Application-Relevant Settings in HYDRA

Maintenance of the HYDRA distribution model - inbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing:

Name of the parameter	Value
To process production orders	
Message type	PPCC2RECORDER
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE =< MLE variant to be used >
Description	PP-PDC – Download of production orders
Log. Target system	Created logical system
Storage duration	10
To process the upload request	
Message type	PPCC2REQCONF
Priority	High
Command	hysapupl.scr
Command parameter	/UPLSEGNAM=E2BP_PP_TIMETICKET
Description	PP-PDC – Upload request
Log. Target system	Created logical system
Storage duration	10
To process variances	
Message type	DIFFE2

Name of the parameter	Value
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE =< MLE variant to be used >
Description	PP-PDC – Variances
Log. Target system	Created logical system
Storage duration	10
To process general quantity units	
Message type	UNIT2
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE =< MLE variant to be used >
Description	PP-PDC – Gen. qty. unit
Log. Target system	Created logical system
Storage duration	10
To process material-dependent quantity units	
Message type	UNIMA2
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE =< MLE variant to be used >
Description	PP-PDC – Gen. qty. unit
Log. Target system	Created logical system
Storage duration	10

Maintenance of the HYDRA distribution model - outbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA outbound processing:

Name of the parameter	Value
To upload time tickets	
Message type	PPCC2PRETTICKET
Description	PP-PDC – Upload of time tickets
IDoc type	PPCC2PRETTICKET01
Storage duration	10
Log. target system	Created logical system
Segment name 1	E2BP_PP_TIMETICKET

Activation of initial download

Since program version

.lib\b_anr.dll V8.1.1.326

the initial download function needs to be enabled explicitly for security reasons.

Create the following entry in the [HYDRA INI configuration](#) if you would like to enable the initial download function for the system:

Parameter name	Value
INI name	SAP
Section	INITIAL_DOWNLOAD_ACTIVATION
Key	ACTIV_TILL

Parameter name	Value
Value	<date in the format MM/DD/YYYY>
Active	Yes
Comment	Activation of the initial download

Behavior when deleting interrupted operations

Specific application functions of the production order or its operations result in a deletion download from SAP to HYDRA. This includes, among other things:

- Setting of a deletion flag
- Technical completion in SAP

By default, data of an interrupted operation will be deleted if the deletion download arrives in HYDRA.

However, special customizing settings within [order status assignment](#) can prevent this process. The following configurations have to be set e.g. for the “interrupted” operation status:

Field	Value
Alterable order data	J or M
Action	E or X

12 Application-Relevant Settings in SAP

Customizing the order type

In SAP the PP-PDC interface will only take those production orders into account, for which the order type has been marked as "BDE active". This is specified in Customizing *SPRO* → *Production* → *Production control* → *Master data* → *Order* → *Order type dependent parameters* (OPL8).

For each relevant combination of plant and order type, the indicator "BDE active" must be set on the "realization" tab.

Definition of new subsystem groupings

To the extent that the subsystem groupings included in the SAP delivery do not suffice, it is possible to define new one using SAP Customizing - *SPRO* → *Labor time management* → *Shop floor data collection* → *General settings* → *Groupings for subsystem connection*.

Maintenance at the workplace

Once an order type is identified as "BDE active", the PP-PDC interface will only take those operations into account, for which at least one subsystem grouping is stored to the workplace.

The subsystem grouping at the workplace is maintained using the workplace maintenance (CR02) → Basic data → Subsystems. There, the relevant subsystem can be selected from several stored subsystems.

Setting of the posting time

Depending on the settings in SAP, the PP-PDC interface supports two confirmation/upload scenarios:

- Immediate posting

If the "Immediate posting" indicator is active in Customizing (CI41), time ticket uploads transferred from HYDRA to SAP will immediately be posted. If this posting cannot be made - if for example a production order is being blocked - the confirmations/uploads will stay pre-posted and will be posted during the next posting run.

- Posting using job

If the "Immediate posting" indicator is not set in Customizing (CI41), the confirmations/uploads will be pre-posted (AFRPO table). They will then be posted later depending on the job, using Job CORUPROC1.

Definition of user fields

The PP-PDC interface can be used to transfer selected user fields of the work plan/ operation from SAP. HYDRA offers default interpretations for these fields. In order to enable the values to be stored to the user fields in SAP, a user field key must be defined and saved to the operation's work plan.

The user field key is defined in Customizing via SPRO → Production → Production control → Master data → Work plan data → User field definition (OPEC).

A meaning must be saved for the following user fields:

SAP user field	SAP user field in the download structure	Meaning
USR00	USERFIELD_CH20_1	Target cycle
USR01	USERFIELD_CH20_2	Te/ tr
USR04	USERFIELD_QUAN	Partitioning

The created user field key and the corresponding values must be stored to the operation's work plan.

Maintenance of the SAP partner agreement – outbound processing

Maintain the following settings for outbound processing in the partner agreement in SAP (WE20)

Parameter name	Value
To download the production orders	
Partner number	Created logical system
Partner type	LS
Message type	PPCC2RECORDER
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	PPCC2RECORDER

To download the upload request	
Partner number	Created logical system
Partner type	LS
Message type	PPCC2REQCONF
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	PPCC2REQCONF01
To download the workplaces	
Partner number	Created logical system
Partner type	LS
Message type	PPCC2RECWORKCENTER
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	PPCC2RECWORKCENTER01
To download the variances	
Partner number	Created logical system
Partner type	LS
Message type	DIFFE2
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately

Parameter name	Value
Basis type	DIFFE2
To download generally applicable quantity units	
Partner number	Created logical system
Partner type	LS
Message type	UNIT2
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	UNIT2
To download material-dependent quantity units	
Partner number	Created logical system
Partner type	LS
Message type	UNIMA2
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	UNIMA2

Maintenance of the SAP partner agreement – inbound processing

Maintain the following settings for inbound processing in the partner agreement in SAP (WE20)

Parameter name	Value
Partner number	Created logical system

Parameter name	Value
Partner type	LS
Message type	PPCC2PRETTICKET
Transaction code	BAPI

Maintenance of the SAP distribution model - outbound processing

Parameter name	Value
To download the production orders	
Model view	Created model view
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Object name/ interface	RCVPRORDCF
Method	ReceiveProdOrder
Filter	If necessary, maintain the BDE grouping as filter criterion
To download the upload request	
Model view	Created model view
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Object name/ interface	RCVPRORDCF
Method	RequestProdOrdConf
To download the workplaces	
Model view	Created model view

Parameter name	Value
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Object name/ interface	RCVPRORDCF
Method	ReceiveWorkCenter
Filter	If necessary, maintain the BDE grouping as filter criterion
To download the variances	
Model view	Created model view
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Message type	DIFFE2
To download generally applicable quantity units	
Model view	Created model view
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Message type	UNIT2
To download material-dependent quantity units	
Model view	Created model view
Sender/ client	Logical system of the client
Recipient/ server	Logical system for the recipient system
Message type	UNIMA2

Maintenance of the SAP distribution model - inbound processing

Parameter name	Value
To upload time tickets	
Model view	Created model view
Sender/ client	Logical system for the sender system
Recipient/ server	Logical system of the client
Object name/ interface	ProdOrdConfirmation
Method	CreatePredefTimeTicketMultiple
Filter	If necessary, maintain the BDE grouping as filter criterion

Planning of relevant jobs

The following programs/ reports must be planned as job to ensure that the PP-PDC interface will operate automatically:

Program/ report	Meaning	Please note:
CIBDOP_DOWN_PP	Download production orders/ operations	Planning of a variant WITHOUT indication of a target system
CIBDCONF_REQUEST	Download of the upload request	Planning of a variant and indicator "Request upload of time tickets" set and as option indication of a target system.

Relevant transactions

Transaction	Meaning	Please note:
CI42N	Download Production orders/ operations	-

CI45N	Download of the upload request	-
CO16N	Reworking of incorrect postings	-

13 Configuration when using SAP PI / SAP PO

Configuration of upload requests in SAP

Assign the fields as described below to configure the job requesting uploads for the PP-PDC interface (SAP transaction CI45N):

Parameter name	Value
Logical system	Empty or logical system created for HYDRA
- Request time ticket upload/confirmation	Leave empty / do not check
Request time ticket events	Leave empty / do not check

Assign the fields as described above if you carry out the transaction manually.

Changing the program call for myerprck.exe/out

When using PI/PO, you have to change the program call for the upload program *myerprck.exe/out* in the [HYDRA Scheduler](#):

Situation	Value
Default program call (Windows) as delivered	sh.exe ./myerprck.scr /MESTYP=PPCC2PRETTICKET /KAT=FA
Call including PI/PO (Windows)	sh.exe ./myerprck.scr /MESTYP=PPCC2PRETTICKET /KAT=FA /PI
Default program call (Linux) as delivered	./myerprck.scr /MESTYP=PPCC2PRETTICKET /KAT=FA > /dev/null 2> /dev/null
Call including PI/PO (Linux)	./myerprck.scr /MESTYP=PPCC2PRETTICKET /KAT=FA /PI > /dev/null 2> /dev/null

Changing the segment name in the distribution model

Create the entry in the [HYDRA distribution model](#) in order to transfer time tickets as outbound configuration based on the following values:

Parameter name	Value
Message type	PPCC2PRETTICKET
Description	PP-PDC – Upload of SAP time tickets
IDoc type	PPCC2PRETTICKET01
Retention period	10
Log. target system	Created logical system
Segment name 1	E2BP_PP_TIMETICKET 000

Create the entry in the HYDRA distribution model in order to request the upload as inbound message type based on the following values:

Parameter name	Value
Message type	PPCC2REQCONF
Priority	High
Command	hysapupl.scr
Command parameter	/UPLSEGNAM=E2BP_PP_TIMETICKET 000
Description	PP-PDC – Upload request
Log. target system	Created logical system
Retention period	10

Requirements

Procedure

Result

14 Protecting fields of planned operations

Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

Procedure from service pack 12 onwards (b_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	The value includes a condition. Enter the condition, for example, as follows: ANR.ATYP=AG
Active	Yes

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value.
Active	Yes

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

Database field	Values/content
INI_VERWEIS	The value of the VERWEIS column identified from the HYD_INI table via SQL.
SECTION	Section
IDENT	Key
VALUE	Value
BEMERKUNG (comment)	Comment
AKTIV	Active

List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

Acronym	Meaning
ANR.MGRP	Machine group
ANR.MNR	Workplace/ machine
ANR.OPT:PLAN	Planning indicator: M Planned for workplace/machine G Planned for machine group
ANR.DATB	Start date planned (via HLS)
ANR.ZEIB	Start time planned (via HLS)

Acronym	Meaning
ANR.DATE	End date planned (via HLS)
ANR.ZEIE	End time planned (via HLS)

Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```


15 Modification to the Order of Uploads

Usage

Subject to the ERP system's way of processing, problems might occur when uploading time tickets to the ERP system. Issue taken: Operations logged on after a shift change are transferred prior to operations interrupted before the change of shifts.

Requirements

The upload program myerprck.exe/out as of version V8.1.1.99 or higher must be used.

Procedure

Add the following entry to the [HYDRA INI configuration](#):

Parameter name	Value
INI name	SAP
Section	TIMETICKET_UPLOAD
Key	ALTERNATIVE_ORDER_BY_CLAUSE
Value	<must remain empty>
Active	Yes
Comment	Changed sorting order

Result

Once activated, uploads will be sorted and reported as follows:

Sort sequence	Comment	Internal number ³
Logoff date		3

³ Internal number refers to numbering used in the SQL program logic. The number is not relevant to the end user.

Sort sequence	Comment	Internal number ³
Logoff time		4
Upload type (record type)	The following order applies for uploading record types: 1. operation logons 2. cancelled operation logoffs 3. cancelled partial uploads, interruptions to operations, batch records (H records) 4. Partial uploads 5. Staff records (B records) 6. interrupted operations, operation logoffs, batch records (H records)	1
Reference		5
Order number		6